

Commerce XI Maths App

Theory (Pages 24–26)

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 2, Unit 2 - Measures of Dispersion | Total questions: 36

Question 1 - MCQ · PG-24 · TH-Q1

A quantity whose value remains unchanged is called a

Options

- A. constant
- B. variable
- C. frequency
- D. deviation

Correct answer: constant

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives constant.

Question 2 - MCQ · PG-24 · TH-Q2

A quantity that can take different values is called a

Options

- A. variable
- B. constant
- C. class boundary
- D. coefficient

Correct answer: variable

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives variable.

Question 3 - MCQ · PG-24 · TH-Q3

The arithmetic mean is a measure of

Options

- A. central tendency
- B. dispersion only
- C. skewness only
- D. class width

Correct answer: central tendency

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives central tendency.

Question 4 - MCQ · PG-24 · TH-Q4

Quartiles divide an ordered data set into

Options

- A. four equal parts
- B. two equal parts
- C. ten equal parts
- D. one hundred equal parts

Correct answer: four equal parts

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives four equal parts.

Question 5 - MCQ · PG-24 · TH-Q5

An average alone cannot describe how observations are

Options

- A. scattered around it
- B. alphabetically arranged
- C. named
- D. rounded

Correct answer: scattered around it

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives scattered around it.

Question 6 - MCQ · PG-24 · TH-Q6

The amount by which observations deviate from an average is called

Options

- A. dispersion
- B. correlation
- C. classification
- D. tabulation

Correct answer: dispersion

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives dispersion.

Question 7 - MCQ · PG-24 · TH-Q7

Dispersion describes the degree to which numerical data tend to

Options

- A. spread about an average value
- B. equal the average
- C. form equal classes
- D. become constant

Correct answer: spread about an average value

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives spread about an average value.

Question 8 - MCQ · PG-24 · TH-Q8

Two series can have the same mean but

Options

- A. different dispersion
- B. the same observations
- C. different sample sizes only
- D. no variation

Correct answer: different dispersion

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives different dispersion.

Question 9 - MCQ · PG-24 · TH-Q9

For comparing the scatter of two series, both series may first be plotted on a

Options

- A. number line
- B. pie chart only
- C. map
- D. calendar

Correct answer: number line

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives number line.

Question 10 - MCQ · PG-25 · TH-Q10

Which is an absolute measure of dispersion?

Options

- A. range
- B. coefficient of variation
- C. percentage
- D. ratio only

Correct answer: range

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives range.

Question 11 - MCQ · PG-25 · TH-Q11

The commonly used measures include range, quartile deviation, variance and

Options

- A. standard deviation
- B. median only
- C. mode only
- D. class mark

Correct answer: standard deviation

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives standard deviation.

Question 12 - MCQ · PG-25 · TH-Q12

Range is defined as

Options

- A. largest value – smallest value
- B. largest value + smallest value
- C. mean – median
- D. $Q_3 - Q_1$ divided by 2

Correct answer: largest value – smallest value

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives largest value – smallest value.

Question 13 - MCQ · PG-25 · TH-Q13

If L is the largest value and S the smallest, range equals

Options

- A. $L - S$
- B. $L + S$
- C. $S - L$
- D. LS

Correct answer: $L - S$

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives $L - S$.

Question 14 - MCQ · PG-25 · TH-Q14

Range is the

Options

- A. simplest measure of dispersion
- B. most detailed frequency table
- C. measure of central tendency
- D. same as variance

Correct answer: simplest measure of dispersion

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives simplest measure of dispersion.

Question 15 - MCQ · PG-25 · TH-Q15

For weights 70, 62, 38, 55, 43, 73, 36, 58, 65, 47, the smallest value is

Options

- A. 36
- B. 38
- C. 43
- D. 47

Correct answer: 36

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 36.

Question 16 - MCQ · PG-25 · TH-Q16

For weights 70, 62, 38, 55, 43, 73, 36, 58, 65, 47, the largest value is

Options

- A. 73
- B. 70
- C. 65
- D. 58

Correct answer: 73

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 73.

Question 17 - MCQ · PG-25 · TH-Q17

For weights 70, 62, 38, 55, 43, 73, 36, 58, 65, 47, the range is

Options

- A. 37
- B. 35
- C. 36
- D. 39

Correct answer: 37

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 37.

Question 18 - MCQ · PG-25 · TH-Q18

Salary (₹00)	30-50	50-70	70-90	90-110	110-130	130-150
No. of employees	7	15	30	24	18	11

For the grouped salary data, the lower limit used for the range is

Options

- A. 30
- B. 50
- C. 70
- D. 150

Correct answer: 30

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 30.

Question 19 - MCQ · PG-25 · TH-Q19

Salary (₹00)	30-50	50-70	70-90	90-110	110-130	130-150
No. of employees	7	15	30	24	18	11

For the grouped salary data, the upper limit used for the range is

Options

- A. 150
- B. 130
- C. 110
- D. 90

Correct answer: 150

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 150.

Question 20 - MCQ · PG-25 · TH-Q20

Salary (₹00)	30-50	50-70	70-90	90-110	110-130	130-150
No. of employees	7	15	30	24	18	11

The range of salaries is

Options

- A. 120
- B. 100
- C. 130
- D. 180

Correct answer: 120

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 120.

Question 21 - MCQ · PG-25 · TH-Q21

Range may be unsuitable when a data set contains

Options

- A. extreme values or outliers
- B. equal values
- C. a mean
- D. frequencies

Correct answer: extreme values or outliers

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives extreme values or outliers.

Question 22 - MCQ · PG-25 · TH-Q22

Quartile deviation is also called

Options

- A. semi-interquartile range
- B. total range
- C. mean deviation
- D. coefficient of variation

Correct answer: semi-interquartile range

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives semi-interquartile range.

Question 23 - MCQ · PG-25 · TH-Q23

Interquartile range equals

Options

- A. $Q_3 - Q_1$
- B. $Q_3 + Q_1$
- C. $Q_2 - Q_1$
- D. $(Q_3 - Q_1)/(2)$

Correct answer: $Q_3 - Q_1$

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives $Q_3 - Q_1$.

Question 24 - MCQ · PG-25 · TH-Q24

Quartile deviation equals

Options

- A. $(Q_3 - Q_1)/(2)$
- B. $Q_3 - Q_1$
- C. $(Q_3 + Q_1)/(2)$
- D. $(Q_2)/(2)$

Correct answer: $(Q_3 - Q_1)/(2)$

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives $(Q_3 - Q_1)/(2)$.

Question 25 - MCQ · PG-25 · TH-Q25

Quartile deviation reduces the influence of

Options

- A. extreme values
- B. the central 50% values
- C. class widths
- D. all frequencies

Correct answer: extreme values

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives extreme values.

Question 26 - MCQ · PG-25–26 · TH-Q26

For the ordered marks 52, 56, 58, 60, 61, 67, 72, 75, 79, 84, 91, Q_1 is

Options

- A. 58
- B. 56
- C. 60
- D. 61

Correct answer: 58

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 58.

Question 27 - MCQ · PG-25–26 · TH-Q27

For the ordered marks 52, 56, 58, 60, 61, 67, 72, 75, 79, 84, 91, Q_3 is

Options

- A. 79
- B. 72
- C. 75
- D. 84

Correct answer: 79

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 79.

Question 28 - MCQ · PG-25–26 · TH-Q28

For the ordered marks 52, 56, 58, 60, 61, 67, 72, 75, 79, 84, 91, the interquartile range is

Options

- A. 21
- B. 10.5
- C. 19
- D. 27

Correct answer: 21

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 21.

Question 29 - MCQ · PG-25–26 · TH-Q29

For the ordered marks 52, 56, 58, 60, 61, 67, 72, 75, 79, 84, 91, the quartile deviation is

Options

- A. 10.5
- B. 21
- C. 9.5
- D. 11

Correct answer: 10.5

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 10.5.

Question 30 - MCQ · PG-26 · TH-Q30

For an open-end frequency distribution, an appropriate measure of dispersion is

Options

- A. quartile deviation
- B. range
- C. largest value
- D. smallest value

Correct answer: quartile deviation

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives quartile deviation.

Question 31 - MCQ · PG-26 · TH-Q31

Income (₹)	Less than 50	50-70	70-90	90-110	110-130	130-150	Above 150
No. of persons	59	102	125	330	200	132	52

[[line:The total frequency N is]]

Options

- A. 1000
- B. 948
- C. 816
- D. 1052

Correct answer: 1000

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 1000.

Question 32 - MCQ · PG-26 · TH-Q32

Income (₹)	Less than 50	50-70	70-90	90-110	110-130	130-150	Above 150
No. of persons	59	102	125	330	200	132	52

The first quartile position is

Options

- A. 250th observation
- B. 125th observation
- C. 500th observation
- D. 750th observation

Correct answer: 250th observation

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 250th observation.

Question 33 - MCQ · PG-26 · TH-Q33

Income (₹)	Less than 50	50-70	70-90	90-110	110-130	130-150	Above 150
No. of persons	59	102	125	330	200	132	52

The first quartile class is

Options

A. 70-90

B. 50-70

C. 90-110

D. 110-130

Correct answer: 70-90

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 70-90.

Question 34 - MCQ · PG-26 · TH-Q34

Income (₹)	Less than 50	50-70	70-90	90-110	110-130	130-150	Above 150
No. of persons	59	102	125	330	200	132	52

The third quartile class is

Options

A. 110-130

B. 90-110

C. 130-150

D. 70-90

Correct answer: 110-130

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 110-130.

Question 35 - MCQ · PG-26 · TH-Q35

Income (₹)	Less than 50	50-70	70-90	90-110	110-130	130-150	Above 150
No. of persons	59	102	125	330	200	132	52

Q_1 is approximately

Options

A. 84.24

B. 70

C. 89

D. 110

Correct answer: 84.24

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 84.24.

Question 36 - MCQ · PG-26 · TH-Q36

Income (₹)	Less than 50	50-70	70-90	90-110	110-130	130-150	Above 150
No. of persons	59	102	125	330	200	132	52

Q_3 is approximately

Options

A. 123.4

B. 110

C. 130

D. 84.24

Correct answer: 123.4

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 123.4.